Week 5 Graphical Lasso

Problem: estimate the covariance in an unknown Gaussian distribution $\mathcal{N}(0, \Sigma)$. $\Sigma \in \mathbb{R}^{p \times p}$

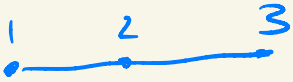
Precision matrix $\Theta := \Sigma^{-1}$

Define undirected graph (V, E)

$V = \{1, \cdots, p\}$

$E \subseteq V \times V$

$(u, v) \notin E \iff X_u \perp X_v \mid X_{V \setminus \{u,v\}}$



$X_1 \perp X_3 \mid X_2$

Hammersley-Clifford theorem

$\Rightarrow \forall (u, v)$

$$X_u \perp X_v \mid X_{V \setminus \{u,v\}} \iff \Theta_{u,v} = 0 \qquad (*)$$

In the Gaussian case, can verify $(*)$ directly.

- iid samples $x^{(1)} \cdots x^{(n)} \sim \mathcal{N}(0, \Theta^{-1})$

log-likelihood:

$$\ell(\Theta) = \sum_{i=1}^{n} \log\left(\det^{1/2}(\Theta)\exp\left(-\frac{1}{2}x^{(i)T}\Theta x^{(i)}\right)\right)$$

$$= \frac{n}{2}\log\det\Theta - \frac{1}{2}\sum_{i=1}^{n} tr\left(x^{(i)}x^{(i)T}\Theta\right)$$

$S := \frac{1}{n}\sum_{i=1}^{n} x^{(i)}x^{(i)T}$

$$= \frac{n}{2}\log\det\Theta - \frac{n}{2}tr(S\Theta)$$

MLE does not exist if S is not full-rank!

$\Rightarrow$ cannot use MLE if $p > n$.

- Graphical-lasso:

$$\underset{\Theta \succeq 0}{\text{minimize}} \quad -\log\det\Theta + tr(S\Theta) + \lambda\|\Theta\|_1 \qquad (\#)$$

$\hookrightarrow$ derivative $= 0$

$$\|\Theta\|_1 := \sum_{i,j}|\Theta_{ij}|$$

$\Rightarrow -\Theta^{-1} + S + \lambda\,\text{sign}(\Theta) = 0$

$\lambda > 0$

$\lambda = 0 \Rightarrow \Theta = S^{-1}$

$(\#)$ is convex opt!

- Dual of graphical lasso:

$$\underset{T:\ \|T\|_\infty \le \lambda}{\text{maximize}} \quad \log\det(S+T) + p \qquad (1)$$

T: symmetric matrix in $\mathbb{R}^{p\times p}$

Claim: $(1) = (\#)$

- Convex conjugate: suppose $\Lambda : \mathbb{R}^p \to \mathbb{R} \cup \{+\infty\}$

Define

$$\Lambda^*(v) := \sup_{u\in\mathbb{R}^p} \{\langle u, v\rangle - \Lambda(u)\} \text{ is convex}$$

$\forall\, v \in \mathbb{R}^p$.

$\Lambda^{**} = (\Lambda^*)^*$ is convex

$\Rightarrow \Lambda$ if Λ is convex (& regularity)

→ Fenchel's duality for $k=2$

Thm: Let $\Lambda_j : \mathbb{R}^p \to \mathbb{R} \cup \{+\infty\}$ convex, $j = 1 \cdots k$.

Regularity conditions (see lecture notes). Then

$$\inf_{v\in\mathbb{R}^p} \left\{ \sum_{j=1}^k \Lambda_j^*(v) \right\} = \sup_{\substack{u_1 \cdots u_k \in \mathbb{R}^p \\ u_1 + \cdots + u_k = 0}} \left\{ -\sum_{j=1}^k \Lambda_j(u_j) \right\}$$

· Check weak duality: $\forall\, v \in \mathbb{R}^p$, $\forall\, u_1, \dots, u_k \in \mathbb{R}^p$, $u_1 + \dots + u_k = 0$

$$\Rightarrow \quad \sum_{j=1}^{k} \Lambda_j^*(v) \geq -\sum_{j=1}^{k} \Lambda_j(u_j)$$

by defn of Λ^*,

$$\sum_{j=1}^{k} \Lambda_j^*(v) \geq \sum_{j=1}^{k} \left\{ \langle u_j, v \rangle - \Lambda_j(u_j) \right\}$$

$$= -\sum_{j=1}^{k} \Lambda_j(u_j)$$

Proof of Claim (1) = (#): want $\forall\, \Theta \in \mathbb{R}^{p^2}$

$$\Lambda_1^*(\Theta) = -\log\det(\Theta)$$

$= +\infty$ if Θ is not p.d.

$$\Lambda_2^*(\Theta) = \operatorname{tr}(S\Theta)$$

$$\Lambda_3^*(\Theta) = \lambda \|\Theta\|_1$$

$\Lambda_1^*(\alpha\Theta_1 + (1-\alpha)\Theta_2)$
$\leq \alpha\Lambda_1^*(\Theta_1) + (1-\alpha)\Lambda_1^*(\Theta_2)$

$$\Lambda_1(T) = \sup_{\Theta \in \mathbb{R}^{p^2}} \left\{ \operatorname{tr}(T\Theta) - \Lambda_1^*(\Theta) \right\}$$

$$= \sup_{\Theta \text{ is p.d.}} \left\{ \operatorname{tr}(T\Theta) + \log\det\Theta \right\}$$

$T + \Theta^{-1} = 0$

$\Theta = -T^{-1}$

$$= -p - \log\det(-T)$$

$$\Lambda_2(T) = \sup_{\Theta \in \mathbb{R}^{p^2}} \left\{ \operatorname{tr}(T\Theta) - \operatorname{tr}(S\Theta) \right\}$$

$$= \begin{cases} 0 & T = S \\ +\infty & \text{o/w} \end{cases}$$

$$\Lambda_3(T) = \sup_{\Theta \in \mathbb{R}^{p^2}} \{ \mathrm{tr}(T\Theta) - \lambda \|\Theta\|_1 \}$$

$$= \begin{cases} 0 & \|T\|_\infty \le \lambda \\ +\infty & \text{o/w} \end{cases}$$

$$\sup_{T_1 + T_2 + T_3 = 0} \{ -\Lambda_1(T_1) - \Lambda_2(T_2) - \Lambda_3(T_3) \}$$

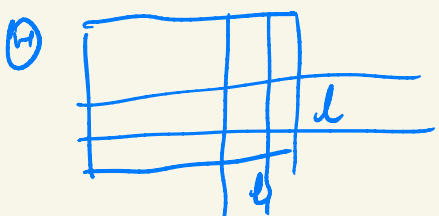
$T_2 = S$, $T_1 = -(S + T_3)$

$$= \sup_{\|T_3\|_\infty \le \lambda} \{ p + \log\det(S + T_3) \}$$

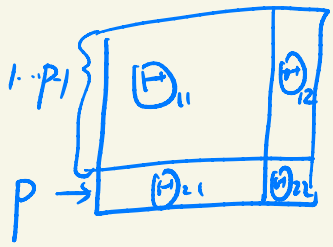
- Blockwise Coordinate Descent

Want: $-\Theta^{-1} + S + \lambda\, \mathrm{sign}(\Theta) = 0$

in each iteration, choose $l \in \{1 \cdots p\}$, optimize l-th column and row



For simplicity, we assume $l = p$



Schur complement theorem

$$\Theta^{-1} = \begin{pmatrix} \cdots & \dfrac{\Theta_{11}^{-1}\Theta_{12}}{\Theta_{22} - \Theta_{21}\Theta_{11}^{-1}\Theta_{12}} \\ \hline \cdots & \dfrac{1}{\Theta_{22} - \Theta_{21}\Theta_{11}^{-1}\Theta_{12}} \end{pmatrix}$$

top right block $\Rightarrow$

$$A\Theta_{12} + S_{12} + \lambda \operatorname{sign}(\Theta_{12}) = 0$$

where $$A := \frac{1}{\Theta_{22} - \Theta_{21}\Theta_{11}^{-1}\Theta_{12}} \left(\Theta_{11}\right)^{-1}$$

Define $$A^{(i-1)} := \frac{1}{\Theta_{22}^{(i-1)} - \Theta_{21}^{(i-1)}\left(\Theta_{11}^{(i-1)}\right)^{-1}\Theta_{12}^{(i)}} \left(\Theta_{11}^{(i-1)}\right)^{-1}$$

Solve $$\underset{\beta \in \mathbb{R}^{p-1}}{\text{minimize}} \quad \frac{1}{2}\beta^T A^{(i-1)}\beta + S_{12}^T\beta + \lambda\|\beta\|_1$$

Set $\Theta_{12}^{(i)} \leftarrow \beta$

lower right block $\Rightarrow$

$$(\Theta^{-1})_{22} = S_{22} + \lambda$$

Compute: $\Theta^{(i)}_{22} = \frac{1}{(\Theta^{-1})_{22}} + \Theta^{(i)}_{21} \left(\Theta^{(i-1)}_{11}\right)^{-1} \Theta^{(i)}_{12}$

Finally, repeat iterations until convergence.

- Note: original graphical lasso by [FHT 08] works for the dual problem instead!